

Lecture Notes

on Hands-On Problems

by Arnaja Mitra

Example: A dose of 250 mg of a medication is taken every 12 hours. The amount of the medication in the bloodstream decays so that every 1.5 hours 85% of the previous amount remains.

1. If the initial amount in the blood is $x_0 = 40$ mg, calculate how much of the medication is in the blood immediately after the fifth dose. (10 pts.)
2. What is the limiting amount of the medication in the blood as time goes on? (5 pts.)

1) let the required difference equation be

$$x_{n+1} = ax_n + b$$

12 hrs.



$$a = ? , b = ?$$

First of all there is $\frac{12}{1.5} = 8$ intervals in each time step of 12 hrs.

$$\therefore a = (0.85)^8 \approx 0.2725.$$

$$b = 250, x_0 = 40.$$

$$x_n = a^n \left[x_0 - \frac{b}{1-a} \right] + \left(\frac{b}{1-a} \right)$$

$$x_5 \approx 343.186 \text{ mg}$$

1 interval (1.5 hrs) decay (0.85)

2nd " $(1.5+1.5)$ " (0.85×0.85)

3rd " " $(0.85)^2$

" " $(0.85)^3$

⋮

2) $\because 0 < a = 0.2725 < 1$, $\therefore$ in a long run the amount of medication in the blood will be

$$\frac{b}{1-a} = \frac{250}{1-0.2725} \approx 343.64 \text{ mg.}$$



Example: A password consists of 3 characters, and each character can be any of the digits 0–9. Digits may repeat. What is the probability that a randomly generated password has all three digits different?

Let S be the sample space of having 3 characters password from 10 digits.

$$\therefore \# S = 10^3.$$

Let E be the event of having password with all three distinct digits.

$$\begin{aligned}\therefore \# E &= {}^{10}P_3 = P(10, 3) = \frac{10!}{(10-3)!} \\ &= \frac{10 \times 9 \times 8 \times \cancel{7!}}{\cancel{7!}} \\ &= 720.\end{aligned}$$

$$\therefore P(E) = \frac{720}{1000} = \dots$$

0 1 2 ... 9

1 3 4

3 1 4

4 1 3

$P(n, r)$

$$= \frac{n!}{(n-r)!}$$

$n = 10, r = 3.$

$$\log_{10} x^k = k \log_{10} x$$

✎ **Example:** Write y as a function of x .

$$\log_{10} y = 0.6 \log_{10} x + 2$$

$$\Rightarrow \log_{10} y = \log_{10} x^{0.6} + 2$$

$$\Rightarrow \log_{10} y = \log_{10} x^{0.6} + \log_{10} 10^2$$

$$\Rightarrow \log_{10} y = \log_{10} (x^{0.6} 10^2)$$

$$\Rightarrow y = 100 x^{0.6}$$

$$a = \log_{10} 10^a$$

$$= a \log_{10} 10 = a$$

$$\log_{10}(\downarrow) = \log_{10}(\downarrow)$$

$$\log_{10} r + \log_{10} s$$

$$= \log_{10} rs$$

$$\log_{10} \left(\frac{r}{s} \right)$$